

PROJECT AETHER-Q // TASK 03

SOVEREIGN ORBITAL QUANTUM TELEMETRY

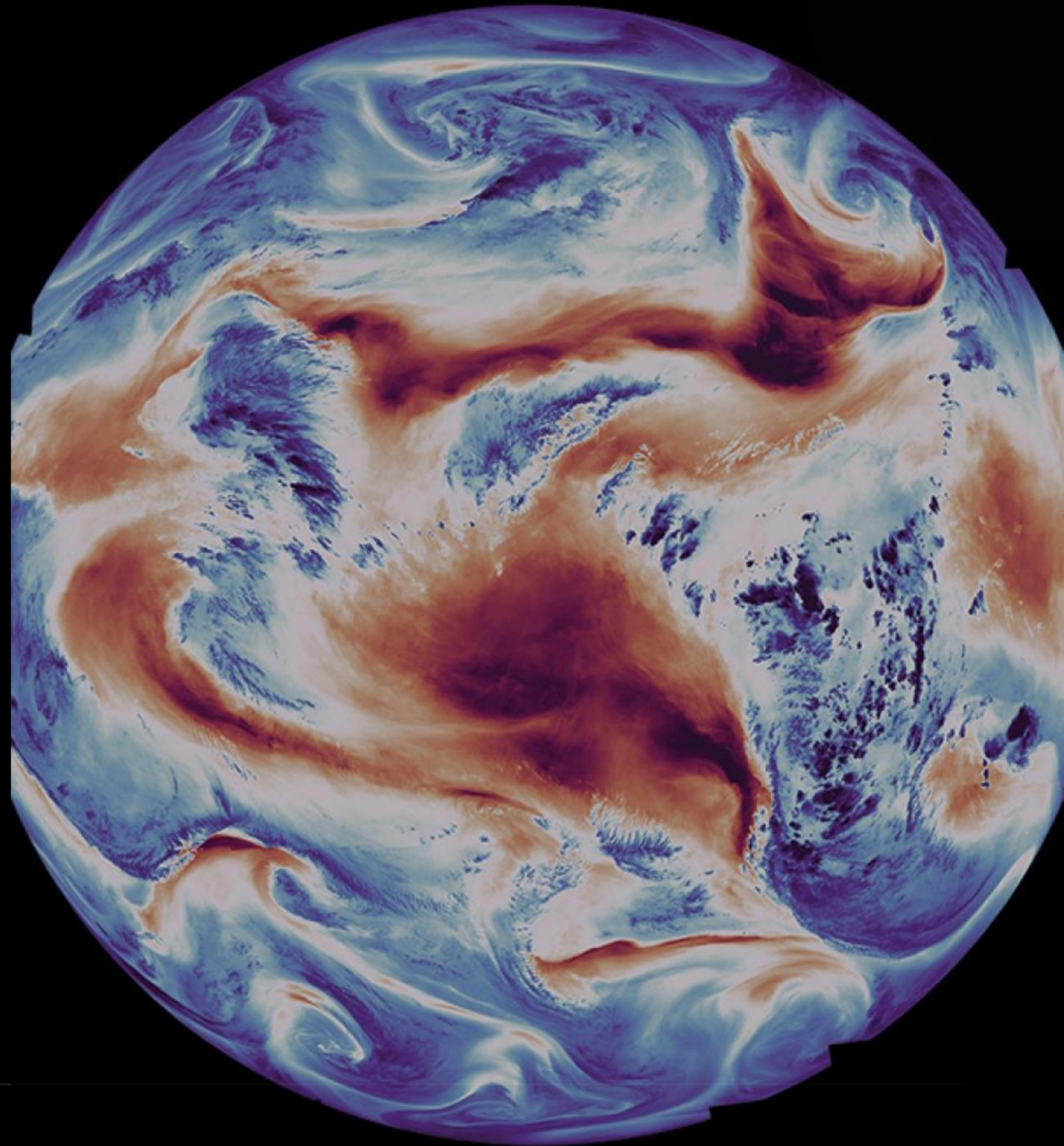
Strategic PRD, Priority Matrix & Execution Roadmap

THE BOTTLENECK

The Orbital Data Downlink Wall

Satellites capture 250+ spectral bands, creating massive raw telemetry files. Downlink contact windows are limited to ~10 mins per orbit.

Transmission costs and power drain make ground-side processing unsustainable for real-time climate monitoring.



| Solution: Quantum Tensor Compression

Hilbert Space Mapping

Project AETHER-Q utilizes Quantum Tensor Neural Networks (QTNNs) to project spectral matrices onto quantum Hilbert spaces.

This achieves **95% data reduction** natively on-orbit before transmission, ensuring 100% data fidelity for anomaly detection.



| Space-Grade Hardware Constraints

8GB

RAM CEILING

Radiation-Hardened Limits

BAE RAD750 and Vorago processors operate at significantly lower clock cycles. Running classical CNNs causes exponential memory scaling $O(2^n)$.

Our QTNN architecture maintains **Linear Scaling** $O(n \cdot m)$, preventing on-orbit thermal shutdown.

[FR-01] Hilbert Hyperspectral Mapper



Ingestion

Real-time ingestion of 250+ spectral bands at ≥ 1.2 Gbps sensor throughput.



Mapping

Project raw data cubes onto compressed quantum state vectors natively on-orbit.



Fidelity

Retain $\geq 99.2\%$ classification accuracy while reducing payload size to $< 5\%$.

| [FR-02] On-Orbit Parametric Classifier

Offline Intelligence

Zero cloud-handshake dependency. Execute pre-compiled PQC operators to detect climate stress anomalies (wildfires, leaks, polar fractures).

Power Efficiency

Total operational draw < 8 Watts. Processes local inference loops under 250ms per spatial swath, staying within LEO thermal envelopes.

| [FR-03] Autonomous Downsampler

- ⚡ **Priority Slicing:** Dynamically identifies anomaly frames for priority ground-station transmission during orbital windows.
 - 📶 **Power-Aware Throttling:** Automatically restricts downlink bandwidth based on real-time solar panel energy reserves.
 - 📁 **Transmission Compiling:** Packages and verifies compressed telemetry files in under 60 seconds of ground station lock.
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RICE Prioritization Matrix

Feature	Reach	Impact	Confidence	Effort	RICE Score
Hilbert Mapper [FR-01]	10.0	3.0	90%	3.5 Mo	7.71
OPC Classifier [FR-02]	8.5	2.5	85%	4.0 Mo	4.51
Autonomous Downsamplers [FR-03]	7.0	1.5	95%	2.5 Mo	3.99

| Execution Roadmap: Phase 1 & 2

Q2: FEATURE MAPPING

Deploy H2IM pipelines to process 250+ spectral bands natively on LEO nodes.

Q1: KERNEL FOUNDATION

Compile base PQC kernels on space test rigs. Validate sub-4W standby power.

Q3: SWARM INTEGRATION

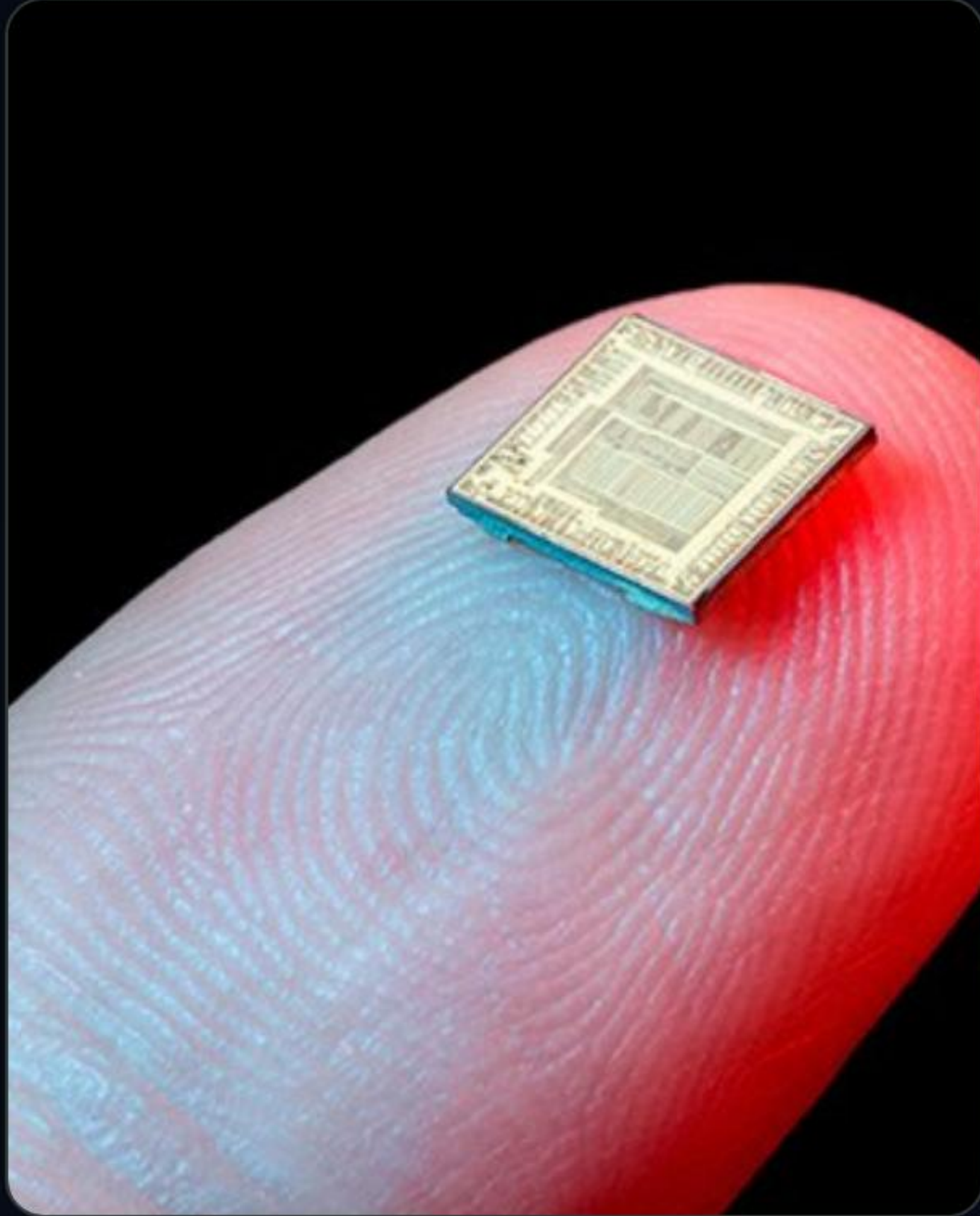
Launch multi-satellite coordination for collaborative terrain feature tracking.

Mission Efficiency Benchmarking



AETHER-Q compresses telemetry footprint by 95% while maintaining lossless data integrity for on-orbit inference.

Market Moats & Defensibility



Hardware Agnosticism

Pre-compiled PQC runtime engine compatible with legacy RAD750 CPUs, bypassing 5-year upgrade cycles.



Sovereign Control

On-orbit encryption prevents ground-relay data sniffing, ensuring absolute planetary security compliance.



Swarm Superiority

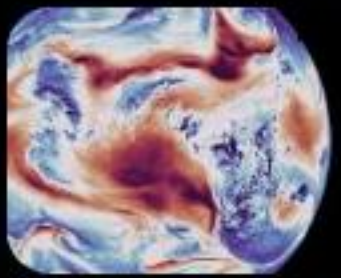
Distributed quantum mesh allows constellations to share compute loads during high-intensity sensor runs.

QUESTIONS?

Project AETHER-Q: Defining the Future of Orbital Intelligence

PORTFOLIO SUCCESS // SKILLCRAFT TECHNOLOGY

| Image Sources



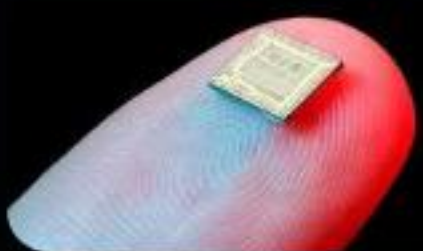
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